Ryan Gupta

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Education

University of Michigan Expected May 2028
B.S. Computer Science and Robotics GPA: 3.4/4.0
Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Probability & Statistics, Linear Algebra, Controls, Cybersecurity, Digital Logic / Verilog

Skills

Programming Languages: Python, C++, Java, C, TypeScript, SQL, MATLAB, Verilog
Frameworks & Tools: Linux, Spring Boot, FastAPI, React, Git, Docker, AWS, OpenCV, ROS 2, MATLAB/Simulink, FAISS, ClickHouse, SQLite, CAN Bus, Jetson

Experience

Visa May 2026 – Aug 2026
Software Engineer Intern *Austin, TX*

- Built a database-backed deduplication system for a multi-tenant monitoring platform, eliminating repeated alerts and redundant LLM triage for identical production failures.
- Built an autonomous incident-response service in Java and Spring Boot using Claude, ClickHouse logs, and operational runbooks to diagnose failures and automatically execute remediation or escalation paths.
- Rebuilt Rubrik deployment and re-enrollment automation across 9 production servers, eliminating 7 observed backup failures and reducing provisioning time from **4 hours** to under **5 minutes**.

Michigan Advanced Robotics Competition (MARC) Mar 2025 – Present
Co-Founder & Software Lead *Ann Arbor, MI*

- Co-founded a competition robotics team, recruiting its initial engineering cohort and coordinating cross-functional development to build a competition-ready robot in 4 months.
- Architecting a C-based embedded control stack using CAN-connected DJI M3508 motors and Jetson compute, with emphasis on deterministic execution, hardware-software interfaces, and real-time control.

University of Michigan Formula SAE Sep 2024 – Apr 2026
Software Engineer *Ann Arbor, MI*

- Cut ROS 2 LiDAR processing latency by 50% and missed obstacles by 42% by profiling C++ and Python nodes, optimizing clustering algorithms, and refining frame-association logic.
- Used test-vehicle telemetry and repeated experiments to validate perception changes end to end before integration into the broader autonomous vehicle software stack.

University of Waterloo Alternative Fuels Team (UWAFT) May 2025 – Aug 2025
Software Engineer *Waterloo, ON*

- Built repeatable MATLAB/Simulink experiments for lane-centering control, enabling systematic comparison of controller settings and vehicle-response metrics across runs.
- Standardized simulation and logging workflows in MATLAB, improving reproducibility and making controller experiments easier to compare across team members.

Projects

Concurrent Delivery System Sep 2026

- Implemented a multithreaded C++20 delivery system with mutex-protected shared state and condition-variable synchronization across concurrent driver and customer threads.
- Coordinated matching between mutually nearest drivers and customers using Manhattan distance, synchronizing delivery and payment handshakes while allowing drivers to operate concurrently.

FinSearch Aug 2026

- Built a financial document search engine indexing **36,599 passages** across 192 SEC 10-K/10-Q filings from 50 public companies using custom BM25 retrieval, FAISS semantic vector search, and reciprocal-rank fusion.
- Built an evaluation harness comparing lexical, semantic, and hybrid search quality and latency, measuring up to **0.893 NDCG@10** and sub-30 ms p95 hybrid retrieval.